Department of Computer Systems Engineering
Faculty of Computing
Sri Lanka Institute of Information Technology

IE2102: Network Programming
Year 2 – Semester II
2026

Assignment Title	Design and Implementation of a Secure Multiprocessor TCP Application Server with Custom Protocol and Session Management
Learning outcomes covered	LO 5: Design and implement a client/server program using socket communication
Assignment Mode	Take-home part and lab assessment part
Maximum Marks	100
Contribution to the Final Grade	25%
Date published	22nd March 2026
Deadline for submissions	12th April 2026
Mode of Submission	Online

Assignment description

1. Environmental Configuration Requirements
    a. Server
        i. Must be implemented in C language
        ii. Must use TCP only
        iii. Must use multiprocessing (`fork`)
        iv. Must compile using `gcc`
        v. Must build using a `Makefile`
        vi. Server File name should be server_<last4digits_registrationNo>.c

            e.g. Assume registration number: IT12345678

            server_5678.c

        vii. Server port number

            PORT = 50 <last 3 digits_registrationNo>

            Example:

            IT12345678 → Port = 50678

        viii. Server ID in Responses

            Every server response must include: SID:<middle4digits>

            e.g. SID:3456

ix. Server Log File Name : server_<regno>.log

server_IT12345678.log

x. Server must store data under: /srv/ie2102/<regno>/<username>/

Srv/ie2102/IT12345678/amal

b. Client
   i. Must be implemented in Python
   ii. Client File name should be client_<last4digits_registrationNo>.py

e.g. Assume registration number: IT12345678

client_5678.py

c. Makefile
   i. Make file name: Makefile_<last4digits>
      Makefile_5678
   ii. Build using: `make -f Makefile_5678`

<u>Assignment Tasks</u>

a. PART A – Take-Home Component

i. A1 – Custom TCP Protocol
   1. Implement explicit framing architecture.
   2. Client sends:

```
LEN:<n>
<payload>
```

   3. Where <n> equals exact payload byte count.
   4. Server must handle:
      a. Partial *recv()*
      b. Multiple messages in one buffer
      c. Invalid length
      d. Oversized payload (>4096 bytes)
   5. Response format:

```
OK <code> SID:<SID><middle4digits> <message>

ERR <code> SID:<SID> <middle4digits><message>
```

ii. A2 – Multiprocessing Design Using `fork()`

6. Server must:
   a. Accept multiple clients concurrently using `fork()`
   b. Parent continues accepting connections
   c. Child handles one client session
7. Proper SIGCHLD handling
   a. `waitpid()` to avoid zombie processes
   b. Clean shutdown behavior
   c. Must demonstrate handling at least 10 concurrent clients

iii. A3 – Authentication + Session Tokens

8. Required commands:

```
REGISTER <user> <pass>

LOGIN <user> <pass>

LOGOUT
```

9. Rules:
   a. Passwords must not be stored in plain text
   b. Must use salted hashing
   c. On login, server issues session token
   d. Protected commands require token
   e. Token expires after inactivity (e.g., 5 minutes)

iv. A4 – Abuse Protection

Must implement:

10. Rate limiting per client
11. Login brute-force lockout
12. Username validation
13. Payload overflow rejection

v. A5 – Persistent Audit Logging

Log must include:

14. Timestamp
15. Client IP:port
16. Child PID
17. Username (if logged in)
18. Command and result
19. Log file name : server_<regno>.log
    e.g. server_IT12345678.log

vi. Create a report for the Take home component.

20. The report must include screenshots of:
    a. Hostname configuration
    b. Network configuration
    c. Server running (port proof using ss/netstat)
    d. Successful REGISTER and LOGIN
    e. Failed login + lockout proof
    f. Multiple forked processes (ps output)
    g. Log file entries
    h. Proper child cleanup (no zombie processes)
21. Screenshots must visibly show your registration number. Provide brief descriptions for each step.
22. Use the CourseWeb front page template as the report cover page.
23. Upload the report before the deadline 12th April 2026 11.59 PM.
24. Report Name : IE2102_Assignment_Report_<regno>.docx
    IE2102_Assingment_Report_IT12345678.docx
25. Submission of a zip file which includes :
    a. Correctly named server file
    b. Correctly named Python client
    c. Log files
    d. Report in word format (.doc or .docx)
    e. ZIP file properly named : IE2102_<regno>.zip

b. PART B – Lab Assessment

i. Students must bring the fully functional Take-Home implementation (PART A) to the lab session.
ii. Lab assignment assessments will informed accordingly via courseweb.
iii. Students MUST attend the allocated lab session on the mentioned date and time according to the schedule which will be published via courseweb.
iv. At the end of the assessment period a viva session will be carried out.
v. Students should be ready to demonstrate any component of the assignment for the evaluations.
vi. Further instructions on the assessment will be given at the lab session.

Marking Scheme

Component	Criteria	Marks	Excellent	Good	Satisfactory	Poor	Very Poor
Part A – Take Home (40 Marks)	**Protocol & Server Implementation (A1 + A2)**	15	Fully correct protocol, handles all edge cases, stable multiprocessing with no zombies	Minor issues in handling or stability	Basic working implementation	Frequent errors	Not working
	Authentication & Security (A3 + A4)	10	Secure hashing, tokens, expiry, rate limit & validation fully implemented	Minor gaps in security features	Basic login works, weak security	Incomplete features	Not implemented
	Logging & System Functionality (A5)	5	Complete logging with all required fields and consistency	Minor missing details	Basic logging present	Poor/inconsistent logs	No logging
	Report (Structure + Evidence)	10	All required screenshots + clear explanations + well-structured	Minor missing elements	Most screenshots present	Weak explanation	Missing/incomplete
Part B – Lab Assessment + Viva (60 Marks)	**System Functionality & Lab Task Implementation**	25	Fully working system + correct lab task implementation	Minor errors	Basic functionality achieved	Incomplete	Not working
	Code Modification & Practical Skills	15	Independently modifies and debugs code efficiently	Minor assistance needed	Moderate understanding	Struggles significantly	Cannot modify
	Viva – Technical Understanding	10	Strong understanding of TCP, fork, protocol, security	Minor gaps	Basic understanding	Weak knowledge	No understanding
	Viva – Explanation & Justification	10	Clearly explains design decisions and system behavior	Minor gaps	Partial explanation	Poor explanation	Cannot explain